

Priyanshu Khandelwal

3rd Year Undergraduate
Department of Computer Application

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Academic Qualifications

Year	Degree/Certificate	Institute
2023 – Present	BCA (AI & ML)	Galgotias University, Greater Noida
2023	CBSE (XII)	St. Gabriels School, Jabalpur
2020	CBSE (X)	Delhi Public School, Jabalpur

Technical Skills

- **Programming Languages:** Python, Flask
- **Software and Libraries:** Jupyter Notebook, Scikit-Learn, Numpy, Pandas, Matplotlib, NLTK, spaCy, JWT

Work Experience

Backend Engineer – AI Infrastructure — MyPetAlly *PostgreSQL, Prisma, OpenRouter, Google Cloud Vision*

- Designed & deployed a scalable AI backend **integrating OCR, LLMs & structured context** for pet healthcare insights.
- Built a **Multi-layer conversational pipeline** with memory management (5K tokens), streaming responses, and system guardrails.
- **Implemented** context compression **Reducing token usage by 42%** and optimizing LLM API **Cost Efficiency**.
- **Developed Emergency Detection** and validation logic with **Production-level reliability** and automated test coverage.

Key Projects

Multispectral Leaf Image analysis (Published in IJEDR, 2026) [📄 Research Paper](#) [📁 GitHub](#) (Dec’25–Jan’26)

Objective	Developed an DL system for plant disease detection & severity estimation using RGB & multispectral Imaging .
Approach	• Trained Model-A (Multispectral CNN) on 5 spectral bands & vegetation indices for severity estimation. • Trained Model-B (RGB CNN) to classify whether a leaf is healthy or diseased from standard RGB images. • Implemented preprocessing pipeline including band stacking, normalization, resizing (128×128), and Performance evaluation using Accuracy, F1-score, ROC-AUC, & Confusion Matrix.
Results	• Demonstrated strong performance in disease detection and severity estimation, highlighting the effectiveness of combining RGB and multispectral deep learning models for precision agriculture.

2D to 3D Img Conversion (Independent Project) [📁 GitHub](#) (Jan’25–Ongoing)

Objective	Estimated Depth from a single 2D image & Visualized it in 3D using Intel-ISL’s MiDaS model.
Approach	• Implemented monocular depth estimation using MiDaS models (DPT_Large, DPT_Hybrid) trained on diverse datasets to predict relative depth from a single RGB image. • Loaded models via torch.hub, applied preprocessing transforms, and generated depth maps • Visualized the output using Open3D for interactive 3D rendering.
Results	• Successfully transformed standard 2D images into rich 3D-like visualizations, enabling applications in robotics, augmented reality, and computer vision research.

AI Data Analysis Agent (Independent Project) [📁 GitHub](#) (Jan’25–Ongoing)

Objective	Developed an AI Agent capable of processing Multiple file formats & delivering interactive data analysis.
Approach	• Extracted data from diverse file types using pandas, docx, fitz, and PIL. • Achieved advanced language understanding & reasoning by implementing the LLaMA-4 API. • Generated insights & visual reports by Analyzing data, Answering queries & Handling follow-ups.
Results	• Delivered meaningful outputs by Extracting data, Analyzing content & Leveraging LLaMA-4 Maverick.

Courses

Python & DSA , Udemy Probability and Statistics, Udemy	Artificial Inteligence, Oracle Machine Learning, Udemy,Stanford university
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Extra-Curricular Activities

- Solved 200+ Leetcode Questions and participated in 7+ Hackathons.
- Selected as Interviewer for 2 internship candidates at MyPetAlly, assessing backend systems and AI fundamentals.
- Participated in Unifest’24, the North India’s Biggest Fest.